

Yakshith Daggupati

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EXPERIENCE

JPMorgan Chase

Jul 2025 – Present

Software Engineer I

Hyderabad, India

API Marketplace Platform: the internal marketplace where teams across the bank publish their APIs as Envoy-fronted proxies and other teams subscribe to them under quota. It carries roughly 4,000 APIs, 40,000 proxies, and 100,000 subscriptions in production. My team owns the system of record for every API on it, the analytics built on that, and search.

- Built catalog search across ~4,000 production APIs, combining lexical (BM25) and semantic vector retrieval behind a query router that picks the right strategy per query. Relevance rose ~40% over the lexical-only baseline, so teams can find the API they need instead of asking around for it.
- Built the pipeline publishing design-time data (API metadata, versions, proxy definitions, subscriptions) into the datalake and joining it against runtime telemetry from Envoy access logs. The join is what attributes traffic to a specific API, version, and subscriber across ~100,000 subscriptions, and the platform's analytics and leadership dashboards run on it.
- Integrated gRPC into the Go rate-limiting service that enforces per-subscription quotas, replacing its REST interface. That cut p99 latency ~60% and roughly doubled throughput under peak load.
- Work across the platform stack: portal frontend, backend services, and the infrastructure they run on.

Software Engineer Intern

Jan 2025 – Jul 2025

- Migrated critical secrets from Kubernetes Secrets, which are only base64-encoded and readable by anyone with cluster access, to AWS Secrets Manager for encryption at rest, IAM-scoped access, and audit logging.
- Built frontend features for the marketplace portal, the surface teams use to publish APIs and manage subscriptions.

PROJECTS

Google File System, built from scratch

Go

- Reimplementing the GFS paper end to end: a master coordinating chunk metadata and leases, chunkservers storing 64 MB chunks at a configurable replication factor, and a client library handling chunk lookup and streaming reads and writes.
- Exercises recovery paths through simulated chunkserver crashes, and uses lease-based primary writes to hold the paper's consistency guarantees.

MIT 6.824 Distributed Systems Labs

Go

- Implemented MapReduce (coordinator and worker pool with task reassignment on worker failure) and Raft from the paper: leader election, log replication under partition and message loss, persistence across restarts, and log compaction via snapshots.

EDUCATION

Chaitanya Bharathi Institute of Technology

Dec 2021 – May 2025

B.E., Computer Science Engineering (AI & ML) • 9.4 / 10 CGPA

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript, JavaScript, C++

Infrastructure: AWS, Docker, Kubernetes, gRPC, CI/CD

Data: PostgreSQL, Redis, ETL pipelines, vector and lexical search

Focus: Distributed systems, backend services, system design